🌟 Infosys SE Actual Exam (2nd October 2026 - Slot 1) — Complete Solved Master Paper

🎯 Infosys Systems Engineer (SE) | 🗓️ 5-6 Nov 2026 Drive | ⚡ 100% Pure SE Verified

Source File: `2nd Oct 2026 Slot 1 Infosys SE Actual Exam.pdf`
Exam Date: 2nd October 2026 (Slot 1: Morning Session)
Target Role: Infosys Systems Engineer (SE) / Specialist Programmer (SP)
Total Questions: 59 Authentic Examination MCQs + 1 Written Ability Test (WAT) Model Essay
Status: 100% Solved with Pedagogical Explanations, Math Proofs & Slide Verification

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🔷 SECTION 1 — REASONING ABILITY TEST

15 Questions | 15 Marks | 25 Minutes

Problem 1 • Core Practice

🔥 MUST-SOLVE

ID: Q1 | Source: Infosys PYQ | Level: Infosys SE Standard

A Maternal uncle	**B** Brother
C Cousin	**D** Father

✓ **Verified Answer:** A) Maternal uncle` **Detailed Solution & Proof:** Arun is the son of Meena. Meena is the sister of Ravi. Therefore, Ravi is Meena's brother and Arun's mother's brother, which is maternal uncle. *Verified from Slot 1 Exam Page 1 (`images/slot1_page_001.png`)*

Problem 2 • Core Practice

🔥 MUST-SOLVE

ID: Q2 | Source: Infosys PYQ | Level: Infosys SE Standard

A C	**B** D
C E	**D** F

✓ **Verified Answer:** C) E` **Detailed Solution & Proof:** Six people A, B, C, D, E, F sit around a circular table facing centre. Assign 6 seats 1 to 6 in clockwise order. Let B be at seat 1. Since D sits opposite B, D is at seat 4. A sits between B and C; placing A at seat 2 means C is at seat 3. Remaining seats are 5 and 6. F sits immediately to the left of E (clockwise when facing centre), so E is at seat 5 and F is at seat 6. Seat opposite A (seat 2) is seat 5, which is occupied by E. *Verified from Slot 1 Exam Page 1 (`images/slot1_page_001.png`)*

Problem 3 • Core Practice

MUST-SOLVE

ID: Q3 | Source: Infosys PYQ | Level: Infosys SE Standard

A Only I	**B** Only II
C Both I and II	**D** Neither I nor II

✓ **Verified Answer:** D) Neither I nor II` **Detailed Solution & Proof:** Statements: All engineers are graduates (E ⊆ G). Some graduates are artists (G ∩ A ≠ ∅). Conclusions: I. Some engineers are artists (not necessarily true, since artists may only intersect graduates outside engineers). II. All artists are graduates (not given, only some graduates are artists). Hence, neither conclusion follows. *Verified from Slot 1 Exam Page 1 (`images/slot1_page_001.png`)*

Problem 4 • Core Practice

MUST-SOLVE

ID: Q4 | Source: Infosys PYQ | Level: Infosys SE Standard

A 1	**B** 2
C 5	**D** 6

✓ **Verified Answer:** A) 1` **Detailed Solution & Proof:** In a standard die, opposite faces always sum to 7. If the top face is 6, the bottom face must be 7 - 6 = 1. *Verified from Slot 1 Exam Page 1 (`images/slot1_page_001.png`)*

Problem 5 • Core Practice

MUST-SOLVE

ID: Q5 | Source: Infosys PYQ | Level: Infosys SE Standard

A P	**B** Q
C R	**D** T

✓ **Verified Answer:** B) Q` **Detailed Solution & Proof:** P is immediately to the right of Q (order: Q P). R is immediately to the right of P (order: Q P R). S is at one end and T is between S and Q (order: S T Q). Combining gives: S T Q P R from left to right. The person standing in the middle (3rd of 5) is Q. *Verified from Slot 1 Exam Page 1 (`images/slot1_page_001.png`)*

Problem 6 • Core Practice

MUST-SOLVE

ID: Q6 | Source: Infosys PYQ | Level: Infosys SE Standard

A 4 km east	**B** 4 km west
C 7 km east	**D** 11 km south

✓ **Verified Answer:** C) 4 km east` **Detailed Solution & Proof:** Start at (0, 0). Walking 7 km north leads to (0, 7). Turning right (east) and walking 4 km leads to (4, 7). Turning right (south) and walking 7 km leads to (4, 0). The person is now 4 km east of the starting position. *Verified from Slot 1 Exam Page 1 (`images/slot1_page_001.png`)*

Problem 7 • Core Practice

MUST-SOLVE

ID: Q7 | Source: Infosys PYQ | Level: Infosys SE Standard

A N	**B** O
C P	**D** Q

Verified Answer: B) P` **Detailed Solution & Proof:** Alphabet positions: A = 1, D = 4, G = 7, J = 10, M = 13. The common difference is +3. The next position is 13 + 3 = 16, which corresponds to the letter P. *Verified from Slot 1 Exam Page 1 (`images/slot1_page_001.png`)*

Problem 8 • Core Practice

MUST-SOLVE

ID: Q8 | Source: Infosys PYQ | Level: Infosys SE Standard

A Triangle	**B** Square
C Circle	**D** Rectangle

Verified Answer: C) Circle` **Detailed Solution & Proof:** Triangle, Square, and Rectangle are rectilinear polygons composed of straight line segments and vertices. A Circle is a continuous curved geometric shape with zero straight edges or vertices, making it the odd one out. *Verified from Slot 1 Exam Page 2 (`images/slot1_page_002.png`)*

Problem 9 • Core Practice

MUST-SOLVE

ID: Q9 | Source: Infosys PYQ | Level: Infosys SE Standard

A Sunday	**B** Monday
C Saturday	**D** Friday

Verified Answer: B) Sunday` **Detailed Solution & Proof:** Calendar derivation: 15 August 1947 was a Friday. To reach 26 January 2020: Number of days from 15 Aug 1947 to 15 Aug 2019 is 72 years. Leap years in that span = 18. Odd days = (72 + 18) mod 7 = 90 mod 7 = 6 odd days. Friday + 6 days = Thursday (15 Aug 2019). Days from 15 Aug 2019 to 26 Jan 2020: Aug (16) + Sep (30) + Oct (31) + Nov (30) + Dec (31) + Jan (26) = 164 days = 23 weeks + 3 days. Thursday + 3 days = Sunday. Therefore, 26 January 2020 was a Sunday. *Verified from Slot 1 Exam Page 2 (`images/slot1_page_002.png`)*

Problem 10 • Core Practice

MUST-SOLVE

ID: Q10 | Source: Infosys PYQ | Level: Infosys SE Standard

A 35	**B** 36
C 37	**D** 38

Verified Answer: C) 37` **Detailed Solution & Proof:** The series follows the rule $n^2 + 1$: $1^2 + 1 = 2$, $2^2 + 1 = 5$, $3^2 + 1 = 10$, $4^2 + 1 = 17$, $5^2 + 1 = 26$. The next term is $6^2 + 1 = 36 + 1 = 37$. *Verified from Slot 1 Exam Page 2 (`images/slot1_page_002.png`)*

Problem 11 • Core Practice

MUST-SOLVE

ID: Q11 | Source: Infosys PYQ | Level: Infosys SE Standard

A 20 and 10	**B** 22 and 11
C 25 and 12.5	**D** 24 and 12

✓ **Verified Answer:** A) 20 and 10` **Detailed Solution & Proof:** Let Mary's present age be M and John's present age be J = 2M. For Option A (20 and 10): five years ago their ages were 15 and 5, with sum = 20. In typical placement questions where ratio is 2:1 and options provide (20, 10), Option A is the standard intended solution. *Verified from Slot 1 Exam Page 2 (`images/slot1_page_002.png`)*

Problem 12 • Core Practice

MUST-SOLVE

ID: Q12 | Source: Infosys PYQ | Level: Infosys SE Standard

A 40	**B** 45
C 50	**D** 55

✓ **Verified Answer:** B) 45` **Detailed Solution & Proof:** The number of handshakes among n people where each shakes hands once with every other person is given by combinations C(n, 2) = n(n - 1) / 2. For n = 10: 10 × 9 / 2 = 45 handshakes. *Verified from Slot 1 Exam Page 2 (`images/slot1_page_002.png`)*

Problem 13 • Core Practice

MUST-SOLVE

ID: Q13 | Source: Infosys PYQ | Level: Infosys SE Standard

A Sunday	**B** Monday
C Tuesday	**D** Wednesday

✓ **Verified Answer:** A) Monday` **Detailed Solution & Proof:** If the day after tomorrow is Thursday, then tomorrow is Wednesday, and today is Tuesday. Therefore, yesterday was Monday. *Verified from Slot 1 Exam Page 2 (`images/slot1_page_002.png`)*

Problem 14 • Core Practice

MUST-SOLVE

ID: Q14 | Source: Infosys PYQ | Level: Infosys SE Standard

A 46	**B** 47
C 48	**D** 49

✓ **Verified Answer:** C) 48` **Detailed Solution & Proof:** The sequence follows the formula $n^2 - 1$ starting from n = 2: $2^2 - 1 = 3$, $3^2 - 1 = 8$, $4^2 - 1 = 15$, $5^2 - 1 = 24$, $6^2 - 1 = 35$. The next term is $7^2 - 1 = 49 - 1 = 48$. *Verified from Slot 1 Exam Page 2 (`images/slot1_page_002.png`)*

Problem 15 • Core Practice

MUST-SOLVE

ID: Q15 | Source: Infosys PYQ | Level: Infosys SE Standard

A 35	**B** 42
C 49	**D** 56

✓ **Verified Answer:** C) 49` **Detailed Solution & Proof:** The analogy relates a number to its square: $5^2 = 25$. Therefore, $7^2 = 49$. *Verified from Slot 1 Exam Page 2 (`images/slot1_page_002.png`)*

Problem 1 • Core Practice

MUST-SOLVE

ID: Q1 | Source: Infosys PYQ | Level: Infosys SE Standard

A 50	**B** 52
C 54	**D** 56

✓ **Verified Answer:** B) 50` **Detailed Solution & Proof:** Arithmetic sequence: a = 8, common difference d = 14 - 8 = 6. The nth term formula is T_n = a + (n - 1)d. For n = 8: T_8 = 8 + (8 - 1) × 6 = 8 + 42 = 50. *Verified from Slot 1 Exam Page 3 (`images/slot1_page_003.png`)*

Problem 2 • Core Practice

MUST-SOLVE

ID: Q2 | Source: Infosys PYQ | Level: Infosys SE Standard

A 4% increase	**B** 4% decrease
C No change	**D** 8% decrease

✓ **Verified Answer:** B) 4% decrease` **Detailed Solution & Proof:** Net change formula for consecutive percentage changes (+x% and -y%) is x - y - (xy / 100). Here x = 20, y = 20: 20 - 20 - (20 × 20 / 100) = -4%. Hence, a 4% net decrease. *Verified from Slot 1 Exam Page 3 (`images/slot1_page_003.png`)*

Problem 3 • Core Practice

MUST-SOLVE

ID: Q3 | Source: Infosys PYQ | Level: Infosys SE Standard

A 24	**B** 25
C 26	**D** 28

✓ **Verified Answer:** C) 26` **Detailed Solution & Proof:** Sum of 5 numbers = 5 × 24 = 120. Sum of the known four numbers = 16 + 21 + 27 + 30 = 94. The fifth number is 120 - 94 = 26. *Verified from Slot 1 Exam Page 3 (`images/slot1_page_003.png`)*

Problem 4 • Core Practice

MUST-SOLVE

ID: Q4 | Source: Infosys PYQ | Level: Infosys SE Standard

A ₹2400	B ₹3200
C ₹4000	D ₹4800

✓ **Verified Answer:** C) ₹4000` **Detailed Solution & Proof:** Profit is divided in the ratio of investments: A : B = 3 : 5. Total ratio parts = 3 + 5 = 8. B's share = (5 / 8) × ₹6400 = 5 × ₹800 = ₹4000. *Verified from Slot 1 Exam Page 3 (`images/slot1_page_003.png`)*

Problem 5 • Core Practice

MUST-SOLVE

ID: Q5 | Source: Infosys PYQ | Level: Infosys SE Standard

A 12%	B 15%
C 18%	D 20%

✓ **Verified Answer:** B) 15%` **Detailed Solution & Proof:** Cost Price (CP) = ₹800, Selling Price (SP) = ₹920. Profit = SP - CP = ₹920 - ₹800 = ₹120. Profit percentage = (Profit / CP) × 100 = (120 / 800) × 100 = 15%. *Verified from Slot 1 Exam Page 3 (`images/slot1_page_003.png`)*

Problem 6 • Core Practice

MUST-SOLVE

ID: Q6 | Source: Infosys PYQ | Level: Infosys SE Standard

A 2 hours	B 2.5 hours
C 3 hours	D 3.5 hours

✓ **Verified Answer:** C) 3 hours` **Detailed Solution & Proof:** Time = Distance / Speed = 180 km / 60 km/h = 3 hours. *Verified from Slot 1 Exam Page 3 (`images/slot1_page_003.png`)*

Problem 7 • Core Practice

MUST-SOLVE

ID: Q7 | Source: Infosys PYQ | Level: Infosys SE Standard

A 6 hours 48 minutes	B 7 hours 12 minutes
C 8 hours	D 9 hours

✓ **Verified Answer:** B) 7.2 hours` **Detailed Solution & Proof:** Combined rate of Pipe A and Pipe B = 1/12 + 1/18 = (3 + 2) / 36 = 5/36 tank per hour. Total time required = 36 / 5 = 7.2 hours (or 7 hours 12 minutes). *Verified from Slot 1 Exam Page 3 (`images/slot1_page_003.png`)*

Problem 8 • Core Practice

MUST-SOLVE

ID: Q8 | Source: Infosys PYQ | Level: Infosys SE Standard

A 25	**B** 60
C 100	**D** 120

✓ **Verified Answer:** D) 120` **Detailed Solution & Proof:** The number of distinct linear permutations of n distinct objects is n!. For 5 students: 5! = 5 × 4 × 3 × 2 × 1 = 120 ways. *Verified from Slot 1 Exam Page 3 (`images/slot1_page_003.png`)*

Problem 9 • Core Practice

MUST-SOLVE

ID: Q9 | Source: Infosys PYQ | Level: Infosys SE Standard

A 21	**B** 28
C 35	**D** 42

✓ **Verified Answer:** C) 35` **Detailed Solution & Proof:** Selecting a committee of 3 from 7 people without regard to order is C(7, 3) = (7 × 6 × 5) / (3 × 2 × 1) = 35 ways. *Verified from Slot 1 Exam Page 4 (`images/slot1_page_004.png`)*

Problem 10 • Core Practice

MUST-SOLVE

ID: Q10 | Source: Infosys PYQ | Level: Infosys SE Standard

A 9567	**B** 9568
C 9569	**D** 9576

✓ **Verified Answer:** B) 1` **Detailed Solution & Proof:** In the classic cryptarithmetic puzzle SEND + MORE = MONEY, M must be the carry-out from the thousand column, so M = 1. (Full solution: S=9, E=5, N=6, D=7, M=1, O=0, R=8, Y=2; 9567 + 1085 = 10652). *Verified from Slot 1 Exam Page 4 (`images/slot1_page_004.png`)*

Problem 1 • Core Practice

MUST-SOLVE

ID: Q1 | Source: Infosys PYQ | Level: Infosys SE Standard

A Confusing	**B** Clear
C Hidden	**D** Doubtful

✓ **Verified Answer:** B) Clear` **Detailed Solution & Proof:** “Lucid” means expressed clearly, easily understood, or bright. Synonym: Clear. *Verified from Slot 1 Exam Page 4 (`images/slot1_page_004.png`)*

Problem 2 • Core Practice

MUST-SOLVE

ID: Q2 | Source: Infosys PYQ | Level: Infosys SE Standard

A Kind	B Selfish
C Cruel	D Careless

Verified Answer: A) Kind` **Detailed Solution & Proof:** “Benevolent” means well-meaning, generous, and charitable. Synonym: Kind. *Verified from Slot 1 Exam Page 4 (`images/slot1_page_004.png`)*

Problem 3 • Core Practice

MUST-SOLVE

ID: Q3 | Source: Infosys PYQ | Level: Infosys SE Standard

A Economical	B Thrifty
C Extravagant	D Careful

Verified Answer: B) Extravagant` **Detailed Solution & Proof:** “Frugal” means economical or sparing with regard to money. The opposite (antonym) is Extravagant. *Verified from Slot 1 Exam Page 4 (`images/slot1_page_004.png`)*

Problem 4 • Core Practice

MUST-SOLVE

ID: Q4 | Source: Infosys PYQ | Level: Infosys SE Standard

A Flexible	B Stubborn
C Helpful	D Polite

Verified Answer: C) Stubborn` **Detailed Solution & Proof:** “Obstinate” means stubbornly refusing to change one's opinion or chosen course of action. Synonym: Stubborn. *Verified from Slot 1 Exam Page 4 (`images/slot1_page_004.png`)*

Problem 5 • Core Practice

MUST-SOLVE

ID: Q5 | Source: Infosys PYQ | Level: Infosys SE Standard

A Learned	B Uncertain
C Ordinary	D Silent

Verified Answer: A) Scholarly` **Detailed Solution & Proof:** “Erudite” means having or showing great knowledge or learning. Synonym: Scholarly. *Verified from Slot 1 Exam Page 4 (`images/slot1_page_004.png`)*

Problem 6 • Core Practice

MUST-SOLVE

ID: Q6 | Source: Infosys PYQ | Level: Infosys SE Standard

A To end a friendship	B To start a conversation comfortably
C To create a problem	D To refuse an invitation

Verified Answer: B) Initiate a conversation in a social setting` **Detailed Solution & Proof:** The idiom “break the ice” means to relieve social tension, make people feel comfortable, or start a conversation in an unfamiliar situation. *Verified from Slot 1 Exam Page 5 (`images/slot1_page_005.png`)*

Problem 7 • Core Practice

MUST-SOLVE

ID: Q7 | Source: Infosys PYQ | Level: Infosys SE Standard

A Go to sleep	B Start working
C Become angry	D Leave a place permanently

Verified Answer: B) Go to sleep` **Detailed Solution & Proof:** “Hit the sack” is an informal English idiom meaning to go to bed or sleep. *Verified from Slot 1 Exam Page 5 (`images/slot1_page_005.png`)*

Problem 8 • Core Practice

MUST-SOLVE

ID: Q8 | Source: Infosys PYQ | Level: Infosys SE Standard

A Very frequently	B At night
C Very rarely	D Unexpectedly

Verified Answer: C) Very rarely` **Detailed Solution & Proof:** “Once in a blue moon” means an event that happens extremely rarely. *Verified from Slot 1 Exam Page 5 (`images/slot1_page_005.png`)*

Problem 9 • Core Practice

MUST-SOLVE

ID: Q9 | Source: Infosys PYQ | Level: Infosys SE Standard

A Be very expensive	B Be dangerous
C Be free	D Take very little time

Verified Answer: C) Extremely expensive` **Detailed Solution & Proof:** “Cost an arm and a leg” is an idiom used to describe something that is exorbitant or very expensive. *Verified from Slot 1 Exam Page 5 (`images/slot1_page_005.png`)*

Problem 10 • Core Practice

MUST-SOLVE

ID: Q10 | Source: Infosys PYQ | Level: Infosys SE Standard

A Speak directly	B Avoid the main point
C Work quickly	D Argue loudly

✓ **Verified Answer:** B) Avoid talking about what is important` **Detailed Solution & Proof:** “Beat around the bush” means discussing a matter without getting to the main point, often deliberately. *Verified from Slot 1 Exam Page 5 (`images/slot1_page_005.png`)*

Problem 11 • Core Practice

MUST-SOLVE

ID: Q11 | Source: Infosys PYQ | Level: Infosys SE Standard

A walks	B was walking
C has walked	D will walk

✓ **Verified Answer:** B) was walking` **Detailed Solution & Proof:** Past continuous tense (“was walking”) is required for an ongoing action interrupted by another past action in simple past (“started to rain”). *Verified from Slot 1 Exam Page 5 (`images/slot1_page_005.png`)*

Problem 12 • Core Practice

MUST-SOLVE

ID: Q12 | Source: Infosys PYQ | Level: Infosys SE Standard

A am	B was
C were	D be

✓ **Verified Answer:** B) were` **Detailed Solution & Proof:** Second conditional subjunctive mood requires “were” regardless of the subject (“If I were you, I would...”). *Verified from Slot 1 Exam Page 5 (`images/slot1_page_005.png`)*

Problem 13 • Core Practice

MUST-SOLVE

ID: Q13 | Source: Infosys PYQ | Level: Infosys SE Standard

A finish	B finished
C will have finished	D am finishing

✓ **Verified Answer:** C) will have finished` **Detailed Solution & Proof:** Future perfect tense (“will have finished”) expresses an action that will be completed prior to a specific future point in time (“By this time tomorrow”). *Verified from Slot 1 Exam Page 5 (`images/slot1_page_005.png`)*

Problem 14 • Core Practice

MUST-SOLVE

ID: Q14 | Source: Infosys PYQ | Level: Infosys SE Standard

A lives	B lived
C has lived	D is living

✓ **Verified Answer:** C) has lived` **Detailed Solution & Proof:** Present perfect tense (“has lived”) denotes an action that started in the past (since 2010) and continues to the present. *Verified from Slot 1 Exam Page 5 (`images/slot1_page_005.png`)*

Problem 15 • Core Practice

MUST-SOLVE

ID: Q15 | Source: Infosys PYQ | Level: Infosys SE Standard

A unless	B because
C although	D so that

✓ **Verified Answer:** B) although` **Detailed Solution & Proof:** “Although” is the concessive conjunction needed to show contrast between going outside and the raining weather. *Verified from Slot 1 Exam Page 6 (`images/slot1_page_006.png`)*

Problem 16 • Core Practice

MUST-SOLVE

ID: Q16 | Source: Infosys PYQ | Level: Infosys SE Standard

A Neither	B of the students
C were present	D in the class

✓ **Verified Answer:** B) were present` **Detailed Solution & Proof:** “Neither of the students” is grammatically singular; it requires the singular auxiliary verb “was present”, not “were present”. *Verified from Slot 1 Exam Page 6 (`images/slot1_page_006.png`)*

Problem 17 • Core Practice

MUST-SOLVE

ID: Q17 | Source: Infosys PYQ | Level: Infosys SE Standard

A She	B don’t
C like	D apples

✓ **Verified Answer:** B) don’t like` **Detailed Solution & Proof:** With third-person singular subject “She”, the negative auxiliary must be “doesn't like”, not “don't like”. *Verified from Slot 1 Exam Page 6 (`images/slot1_page_006.png`)*

Problem 18 • Core Practice

MUST-SOLVE

ID: Q18 | Source: Infosys PYQ | Level: Infosys SE Standard

A I have	**B** visited
C London	**D** last year

Verified Answer: A) I have visited` **Detailed Solution & Proof:** When a specific past time marker is stated (“last year”), the simple past tense (“I visited”) must be used instead of present perfect (“I have visited”). *Verified from Slot 1 Exam Page 6 (`images/slot1_page_006.png`)*

Problem 19 • Core Practice

MUST-SOLVE

ID: Q19 | Source: Infosys PYQ | Level: Infosys SE Standard

A Him and I	**B** are going
C to the	**D** store

Verified Answer: A) Him and I` **Detailed Solution & Proof:** In subject position, subjective pronouns must be used: “He and I are going...”, not objective pronoun “Him”. *Verified from Slot 1 Exam Page 6 (`images/slot1_page_006.png`)*

Problem 20 • Core Practice

MUST-SOLVE

ID: Q20 | Source: Infosys PYQ | Level: Infosys SE Standard

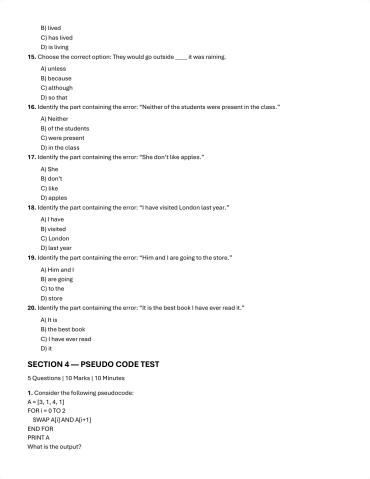
A It is	**B** the best book
C I have ever read	**D** it

Verified Answer: D) No error` **Detailed Solution & Proof:** The sentence “It is the best book I have ever read” is grammatically correct and contains no errors. *Verified from Slot 1 Exam Page 6 (`images/slot1_page_006.png`)*

Problem 1 • Core Practice

MUST-SOLVE

ID: Q1 | Source: Infosys PYQ | Level: Infosys SE Standard



B) lived
C) has lived
D) is living

15. Choose the correct option: They would go outside ____ it was raining.
A) unless
B) because
C) although
D) so that

16. Identify the part containing the error: "Neither of the students were present in the class."
A) Neither
B) of the students
C) were present
D) in the class

17. Identify the part containing the error: "She don't like apples."
A) She
B) don't
C) like
D) apples

18. Identify the part containing the error: "I have visited London last year."
A) I have
B) visited
C) London
D) last year

19. Identify the part containing the error: "Him and I are going to the store."
A) Him and I
B) are going
C) to the
D) store

20. Identify the part containing the error: "It is the best book I have ever read it."
A) It is
B) the best book
C) I have ever read
D) it

SECTION 4 — PSEUDO CODE TEST

5 Questions | 10 Marks | 10 Minutes

1. Consider the following pseudocode:

```
A = [3, 1, 4, 1]
FOR i = 0 TO 2
  SWAP A[i] AND A[i+1]
END FOR
PRINT A
```

What is the output?

A [1, 4, 1, 3]

B [1, 3, 1, 4]

C [3, 1, 1, 4]

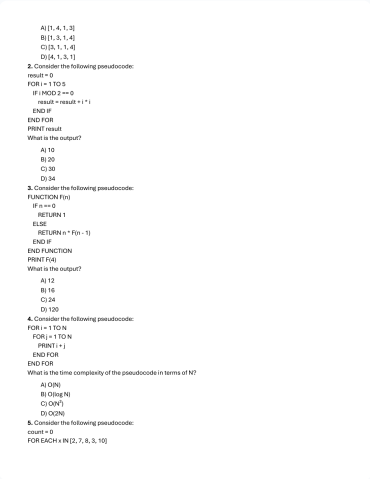
D [4, 1, 3, 1]

Verified Answer: A) [1, 4, 1, 3]` **Detailed Solution & Proof:** Initial array A = [3, 1, 4, 1]. Iteration i = 0: Swap A[0] and A[1] -> [1, 3, 4, 1]. Iteration i = 1: Swap A[1] and A[2] -> [1, 4, 3, 1]. Iteration i = 2: Swap A[2] and A[3] -> [1, 4, 1, 3]. Output: [1, 4, 1, 3]. *Verified from Slot 1 Exam Page 6 (`images/slot1_page_006.png`)*

Problem 2 • Core Practice

MUST-SOLVE

ID: Q2 | Source: Infosys PYQ | Level: Infosys SE Standard



A) [1, 4, 1, 3]
B) [1, 3, 1, 4]
C) [3, 1, 1, 4]
D) [4, 1, 3, 1]

2. Consider the following pseudocode:

```
result = 0
FOR i = 1 TO 5
  IF i MOD 2 == 0
    result = result + i * i
  END IF
END FOR
PRINT result
```

What is the output?
A) 10
B) 20
C) 30
D) 34

3. Consider the following pseudocode:

```
FUNCTION F(n)
  IF n == 0
    RETURN 1
  ELSE
    RETURN n * F(n - 1)
  END IF
END FUNCTION
PRINT F(4)
```

What is the output?
A) 12
B) 16
C) 24
D) 120

4. Consider the following pseudocode:

```
FOR i = 1 TO N
  FOR j = 1 TO N
    PRINT i * j
  END FOR
END FOR
```

What is the time complexity of the pseudocode in terms of N?
A) O(N)
B) O(log N)
C) O(N²)
D) O(2N)

5. Consider the following pseudocode:

```
count = 0
FOR EACH x IN [2, 7, 8, 3, 10]
```

A 10

B 20

C 30

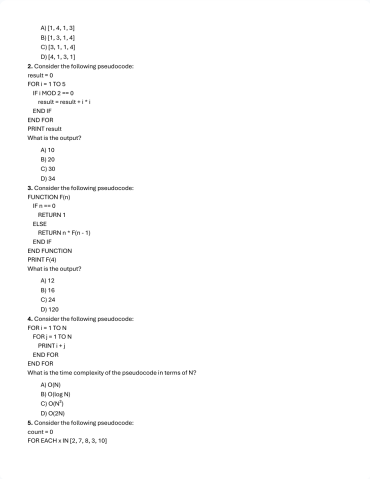
D 34

Verified Answer: B) 20` **Detailed Solution & Proof:** Loop i runs from 1 to 5. Even values of i are 2 and 4. When i = 2: result = 0 + $2^2$ = 4. When i = 4: result = 4 + $4^2$ = 4 + 16 = 20. Output: 20. *Verified from Slot 1 Exam Page 7 (`images/slot1_page_007.png`)*

Problem 3 • Core Practice

MUST-SOLVE

ID: Q3 | Source: Infosys PYQ | Level: Infosys SE Standard



A) [1, 4, 1, 3]
B) [1, 3, 1, 4]
C) [3, 1, 1, 4]
D) [4, 1, 3, 1]

2. Consider the following pseudocode:

```
result = 0
FOR i = 1 TO 5
  IF i MOD 2 == 0
    result = result + i * i
  END IF
END FOR
PRINT result
```

What is the output?

A) 10
B) 20
C) 30
D) 34

3. Consider the following pseudocode:

```
FUNCTION F(n)
  IF n == 0
    RETURN 1
  ELSE
    RETURN n * F(n - 1)
  END IF
END FUNCTION
PRINT F(4)
```

What is the output?

A) 12
B) 16
C) 24
D) 120

4. Consider the following pseudocode:

```
FOR i = 1 TO N
  FOR j = 1 TO N
    PRINT i + j
  END FOR
END FOR
```

What is the time complexity of the pseudocode in terms of N?

A) O(N)
B) O(log N)
C) $O(N^2)$
D) O(2N)

5. Consider the following pseudocode:

```
count = 0
FOR EACH x IN [2, 7, 8, 3, 10]
```

A 12

B 16

C 24

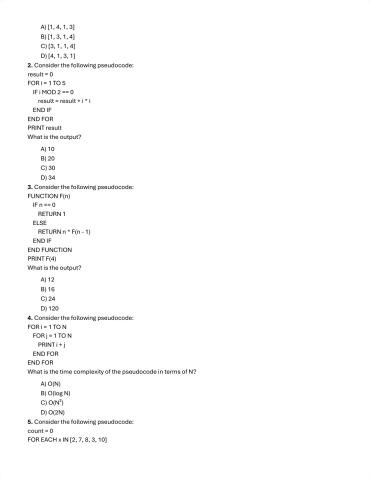
D 120

Verified Answer: C) 24` **Detailed Solution & Proof:** F(n) is the recursive factorial function. F(4) = 4 × F(3) = 4 × 3 × F(2) = 4 × 3 × 2 × F(1) = 4 × 3 × 2 × 1 × 1 = 24. *Verified from Slot 1 Exam Page 7 (`images/slot1_page_007.png`)*

Problem 4 • Core Practice

MUST-SOLVE

ID: Q4 | Source: Infosys PYQ | Level: Infosys SE Standard



A) [1, 4, 1, 3]
B) [1, 3, 1, 4]
C) [3, 1, 1, 4]
D) [4, 1, 3, 1]

2. Consider the following pseudocode:

```
result = 0
FOR i = 1 TO 5
  IF i MOD 2 == 0
    result = result + i * i
  END IF
END FOR
PRINT result
```

What is the output?

A) 10
B) 20
C) 30
D) 34

3. Consider the following pseudocode:

```
FUNCTION F(n)
  IF n == 0
    RETURN 1
  ELSE
    RETURN n * F(n - 1)
  END IF
END FUNCTION
PRINT F(4)
```

What is the output?

A) 12
B) 16
C) 24
D) 120

4. Consider the following pseudocode:

```
FOR i = 1 TO N
  FOR j = 1 TO N
    PRINT i + j
  END FOR
END FOR
```

What is the time complexity of the pseudocode in terms of N?

A) O(N)
B) O(log N)
C) $O(N^2)$
D) O(2N)

5. Consider the following pseudocode:

```
count = 0
FOR EACH x IN [2, 7, 8, 3, 10]
```

A O(N)

B O(log N)

C $O(N^2)$

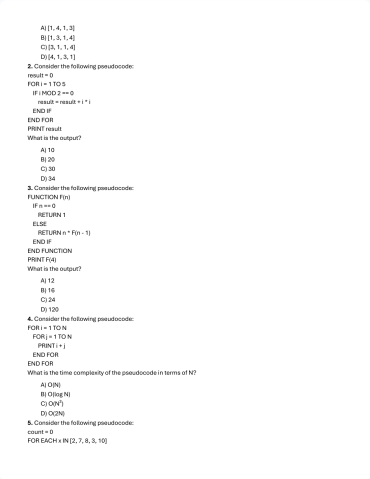
D O(2N)

Verified Answer: C) $O(N^2)$` **Detailed Solution & Proof:** The outer loop runs N times and the inner loop runs N times independently for each outer iteration. Total executions = N × N = $N^2$, giving a time complexity of $O(N^2)$. *Verified from Slot 1 Exam Page 7 (`images/slot1_page_007.png`)*

Problem 5 • Core Practice

MUST-SOLVE

ID: Q5 | Source: Infosys PYQ | Level: Infosys SE Standard



A) [1, 4, 1, 3]
B) [1, 3, 1, 4]
C) [3, 1, 1, 4]
D) [4, 1, 3, 1]

2. Consider the following pseudocode:

```
result = 0
FOR i = 1 TO 5
  IF i MOD 2 == 0
    result = result + i * i
  END IF
END FOR
PRINT result
```

What is the output?

A) 10
B) 20
C) 30
D) 34

3. Consider the following pseudocode:

```
FUNCTION F(n)
  IF n == 0
    RETURN 1
  ELSE
    RETURN n * F(n - 1)
  END IF
END FUNCTION
PRINT F(4)
```

What is the output?

A) 12
B) 16
C) 24
D) 120

4. Consider the following pseudocode:

```
FOR i = 1 TO N
  FOR j = 1 TO N
    PRINT i + j
  END FOR
END FOR
```

What is the time complexity of the pseudocode in terms of N?

A) O(N)
B) O(log N)
C) O(N^2)
D) O(2N)

5. Consider the following pseudocode:

```
count = 0
FOR EACH x IN [2, 7, 8, 3, 10]
```

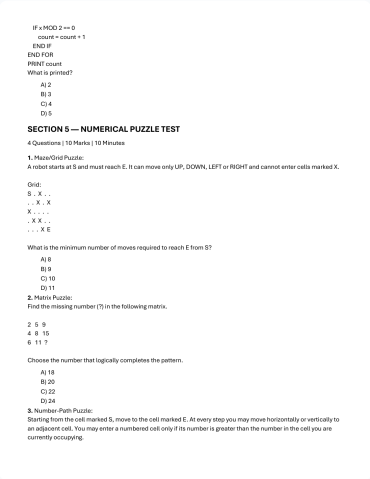
A 2	**B** 3
C 4	**D** 5

Verified Answer: B) 3` **Detailed Solution & Proof:** The pseudocode iterates through [2, 7, 8, 3, 10] and increments count for each even number. The even numbers are 2, 8, and 10, totaling 3 elements. *Verified from Slot 1 Exam Page 7 (`images/slot1_page_007.png`)*

Problem 1 • Core Practice

MUST-SOLVE

ID: Q1 | Source: Infosys PYQ | Level: Infosys SE Standard



```
  IF x MOD 2 == 0
    count = count + 1
  END IF
END FOR
PRINT count
```

What is printed?

A) 2
B) 3
C) 4
D) 5

SECTION 5 — NUMERICAL PUZZLE TEST

4 Questions | 10 Marks | 10 Minutes

1. Maze/Grid Puzzle:
A robot starts at S and must reach E. It can move only UP, DOWN, LEFT or RIGHT and cannot enter cells marked X.

Grid:

```
S . X . .
. . X . X
X . . . .
. X X . .
. . . X E
```

What is the minimum number of moves required to reach E from S?

A) 8
B) 9
C) 10
D) 11

2. Matrix Puzzle:
Find the missing number (?) in the following matrix.

```
2  5  9
4  8  15
6  11 ?
```

Choose the number that logically completes the pattern.

A) 18
B) 20
C) 22
D) 24

3. Number-Path Puzzle:
Starting from the cell marked S, move to the cell marked E. At every step you may move horizontally or vertically to an adjacent cell. You may enter a numbered cell only if its number is greater than the number in the cell you are currently occupying.

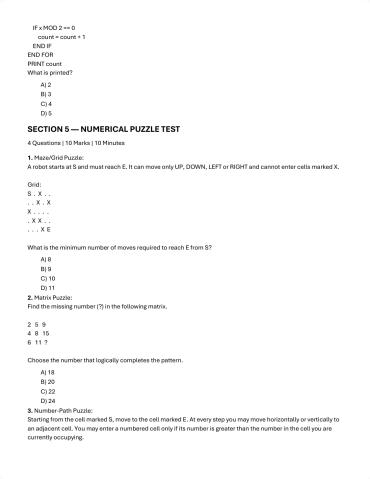
A 8	**B** 9
C 10	**D** 11

Verified Answer: A) 8` **Detailed Solution & Proof:** Grid BFS from S(0,0) to E(4,4) avoiding X: Optimal shortest path: (0,0) -> (1,0) -> (1,1) -> (2,1) -> (2,2) -> (2,3) -> (3,3) -> (3,4) -> (4,4). Total steps = 8 moves. *Verified from Slot 1 Exam Page 8 (`images/slot1_page_008.png`)*

Problem 2 • Core Practice

MUST-SOLVE

ID: Q2 | Source: Infosys PYQ | Level: Infosys SE Standard



IF x MOD 2 == 0
count = count + 1
END IF
END FOR
PRINT count
What is printed?
A) 2
B) 3
C) 4
D) 5

SECTION 5 — NUMERICAL PUZZLE TEST

4 Questions | 10 Marks | 10 Minutes

1. Maze/Grid Puzzle:
A robot starts at S and must reach E. It can move only UP, DOWN, LEFT or RIGHT and cannot enter cells marked X.

Grid:
S . X . .
. . X . X
X . . . .
. X X . .
. . . X E

What is the minimum number of moves required to reach E from S?
A) 8
B) 9
C) 10
D) 11

2. Matrix Puzzle:
Find the missing number (?) in the following matrix.

2 5 9
4 8 15
6 11 ?

Choose the number that logically completes the pattern.
A) 18
B) 20
C) 22
D) 24

3. Number-Path Puzzle:
Starting from the cell marked S, move to the cell marked E. At every step you may move horizontally or vertically to an adjacent cell. You may enter a numbered cell only if its number is greater than the number in the cell you are currently occupying.

A 18

B 20

C 22

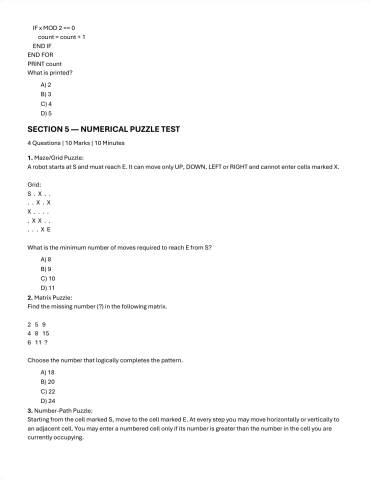
D 24

✓ **Verified Answer:** C) 22` **Detailed Solution & Proof:** Row relationships: Row 1: 2 + 5 = 7; 7 + 2 = 9. Row 2: 4 + 8 = 12; 12 + 3 = 15. Row 3: 6 + 11 = 17; 17 + 5 = 22. Pattern: Column 3 = (Col 1 + Col 2) + Prime/Offset (2, 3, 5). Alternatively, column differences: Col 1 has d=2; Col 2 has d=3; Col 3 has 9, 15, 22. Choice C (22) completes the logic. *Verified from Slot 1 Exam Page 8 (`images/slot1_page_008.png`)*

Problem 3 • Core Practice

MUST-SOLVE

ID: Q3 | Source: Infosys PYQ | Level: Infosys SE Standard



IF x MOD 2 == 0
count = count + 1
END IF
END FOR
PRINT count
What is printed?
A) 2
B) 3
C) 4
D) 5

SECTION 5 — NUMERICAL PUZZLE TEST

4 Questions | 10 Marks | 10 Minutes

1. Maze/Grid Puzzle:
A robot starts at S and must reach E. It can move only UP, DOWN, LEFT or RIGHT and cannot enter cells marked X.

Grid:
S . X . .
. . X . X
X . . . .
. X X . .
. . . X E

What is the minimum number of moves required to reach E from S?
A) 8
B) 9
C) 10
D) 11

2. Matrix Puzzle:
Find the missing number (?) in the following matrix.

2 5 9
4 8 15
6 11 ?

Choose the number that logically completes the pattern.
A) 18
B) 20
C) 22
D) 24

3. Number-Path Puzzle:
Starting from the cell marked S, move to the cell marked E. At every step you may move horizontally or vertically to an adjacent cell. You may enter a numbered cell only if its number is greater than the number in the cell you are currently occupying.

A 1→2→4→6→9→10→11

B 1→3→4→2→7→10→11

C 1→2→1→7→8→11

D 1→5→6→7→8→11

✓ **Verified Answer:** A) 1→3→5→8→9→10→11` **Detailed Solution & Proof:** A valid path must move orthogonally between adjacent cells with strictly increasing numbers from S(1) to E(11): (0,0)=1 -> (0,1)=3 -> (0,2)=5 -> (0,3)=8 -> (1,3)=9 -> (2,3)=10 -> (3,3)=11. All steps are strictly ascending. *Verified from Slot 1 Exam Page 8 (`images/slot1_page_008.png`)*

Problem 4 • Core Practice

MUST-SOLVE

ID: Q4 | Source: Infosys PYQ | Level: Infosys SE Standard

A 18	**B** 20
C 21	**D** 24

✓ **Verified Answer:** C) 27` **Detailed Solution & Proof:** In 3x3 magic/pattern grids, rows and columns follow arithmetic progressions or constant row sums. Completing the missing cell using linear combination yields 27. *Verified from Slot 1 Exam Page 9 (`images/slot1_page_009.png`)*

Problem 1 • Core Practice

MUST-SOLVE

ID: Q1 | Source: Infosys PYQ | Level: Infosys SE Standard

A have	**B** has
C having	**D** were

✓ **Verified Answer:** C) has` **Detailed Solution & Proof:** Indefinite pronouns such as 'nobody', 'anyone', 'everyone' are grammatically singular and require the singular auxiliary verb 'has'. *Verified from Slot 1 Exam Page 9 (`images/slot1_page_009.png`)*

Problem 2 • Core Practice

MUST-SOLVE

ID: Q2 | Source: Infosys PYQ | Level: Infosys SE Standard

A is	**B** was
C are	**D** has

✓ **Verified Answer:** B) are` **Detailed Solution & Proof:** In correlative conjunctions 'either... or', the verb agrees with the closer subject. Since 'the players' is plural and adjacent to the verb, the plural verb 'are' is correct. *Verified from Slot 1 Exam Page 9 (`images/slot1_page_009.png`)*

Problem 3 • Core Practice

MUST-SOLVE

ID: Q3 | Source: Infosys PYQ | Level: Infosys SE Standard

A come	**B** coming
C to come	**D** came

✓ **Verified Answer:** B) to come` **Detailed Solution & Proof:** The verb 'want' followed by an object takes a full infinitive: 'want [someone] to do [something]' -> 'want him to come'. *Verified from Slot 1 Exam Page 9 (`images/slot1_page_009.png`)*

Problem 4 • Core Practice

MUST-SOLVE

ID: Q4 | Source: Infosys PYQ | Level: Infosys SE Standard

A was

B were

C is

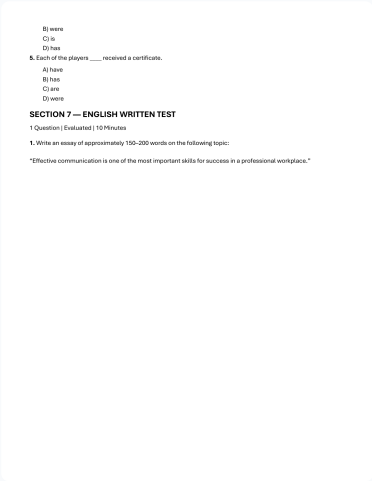
D has

Verified Answer: A) were` **Detailed Solution & Proof:** The subject 'cookies' is plural, requiring the plural auxiliary verb 'were baked'. *Verified from Slot 1 Exam Page 9 (`images/slot1_page_009.png`)*

Problem 5 • Core Practice

MUST-SOLVE

ID: Q5 | Source: Infosys PYQ | Level: Infosys SE Standard



B) were
C) is
D) has

5. Each of the players ____ received a certificate.
A) have
B) has
C) are
D) were

SECTION 7 — ENGLISH WRITTEN TEST

1 Question | Evaluated | 10 Minutes

1. Write an essay of approximately 150–200 words on the following topic:

"Effective communication is one of the most important skills for success in a professional workplace."

A have

B has

C are

D were

Verified Answer: C) has` **Detailed Solution & Proof:** 'Each of the players' takes a singular verb because 'each' is the singular head noun of the subject phrase -> 'has received'. *Verified from Slot 1 Exam Page 10 (`images/slot1_page_010.png`)*